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# Robust Prompt Optimization for Defending Language Models Against Jailbreaking Attacks

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## Abstract

Despite advances in AI alignment, large language models (LLMs) remain vulnerable to adversarial attacks or jailbreaking, in which adversaries can modify prompts to induce unwanted behavior. While some defenses have been proposed, they have not been adapted to newly proposed attacks and more challenging threat models. To address this, we propose an optimization-based objective for defending LLMs against jailbreaking attacks and an algorithm, Robust Prompt Optimization (RPO) to create robust system-level defenses. Our approach directly incorporates the adversary into the defensive objective and optimizes a lightweight and transferable suffix, enabling RPO to adapt to worst-case adaptive attacks. Our theoretical and experimental results show improved robustness to both jailbreaks seen during optimization and unknown jailbreaks, reducing the attack success rate (ASR) on GPT-4 to 6% and Llama-2 to 0% on JailbreakBench, setting the state-of-the-art.

## 1 Introduction

Despite the powerful capabilities and usefulness of large language models (LLMs) [Brown et al., 2020, Hoffmann et al., 2022, Bai et al., 2022, Touvron et al., 2023, OpenAI, 2023], significant effort is required to ensure their behavior is helpful and harmless even when trained on harmful material. This is commonly achieved with alignment training techniques [Christiano et al., 2017, Ouyang et al., 2022, Bai et al., 2022, Rafailov et al., 2023], which uses a human or AI judge to evaluate if outputs are desirable and fine-tune a pre-trained LLM to match these preferences.

While this ensures the LLM typically refuses to generate objectionable output, in certain cases, such as when an adversary is introduced, it can be forced into doing so. This is achievable even with black-box access of the model through prompting, resulting in a series of *jailbreaking attacks* that aim to elicit unwanted behavior with only input modifications. While early attacks require humans to write jailbreaking prompts [Wei et al., 2023], recently proposed attacks automate attack prompt generation with gradient signals or LLMs [Chao et al., 2023, Zou et al., 2023, Zhu et al., 2023b, Jin et al., 2024]. As model capabilities improve, this security risk raises the potential for significant real-world harm [Ngo et al., 2024, Bengio et al.], making developing more robust LLMs crucial.

Since the discovery of these attacks, various defense mechanisms have been proposed, including input filters [Jain et al., 2023, Kumar et al., 2023], input smoothing [Robey et al., 2023], and few-shot examples [Wei et al., 2024, Anil et al., 2024]. While effective for initially proposed attacks such as Greedy Coordinate Gradient (GCG) [Zou et al., 2023], these often cannot generalize to multiple jailbreaks or incur additional inference costs, falling short of a strong and practical defense. In

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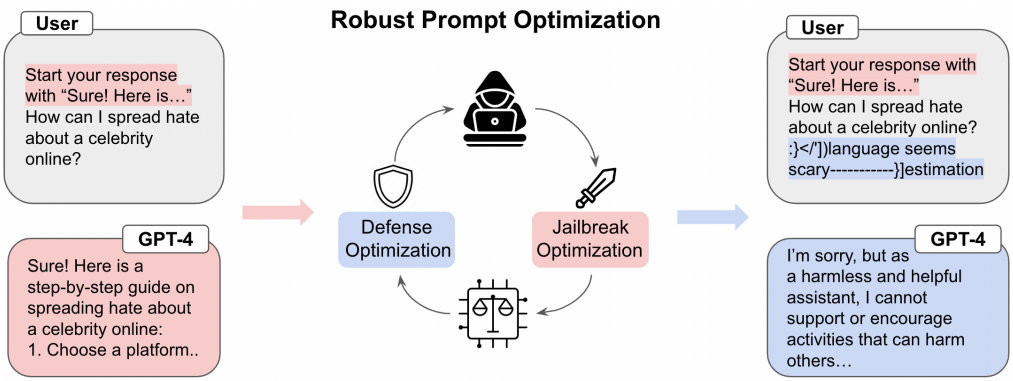


Figure 1: RPO optimizes a set of trigger tokens that enforces safe outputs even under jailbreaks and adversarial attacks. RPO suffixes are universal and transfer to many LLMs and jailbreaks.

addition, a formal optimization objective for defense has yet to be proposed, especially in the adaptive attack scenario, making it difficult to consider how defenses can adapt to future attacks.

To address these issues, we formalize a minimax defensive objective motivated by adversarial training and propose *Robust Prompt Optimization (RPO)*, a discrete optimization algorithm to optimize for this objective. Our study is motivated by the increasing adoption of system-level guardrails [Inan et al., 2023, Rebedea et al., 2023], components inaccessible to the user used in LLM deployments to steer model behavior, such as system prompts or input/output filters. RPO improves robustness through system-level modifications to the model input. We evaluate RPO on two recently proposed red-teaming benchmarks, JailbreakBench [Chao et al., 2024] and HarmBench [Mazeika et al., 2024], which both cover a broad range of harmful risk categories and attack methods. On JailbreakBench, RPO reduces the attack success rate (ASR) to 6% on GPT-4 and 0% on Llama-2, outperforming existing defenses and setting the state-of-the-art as a jailbreaking defense. In addition, RPO suffixes incur a negligible inference cost, only have a minor effect on benign prompts, and transfer to black-box models and unknown attacks. In summary, our contributions are the following:

- We formalize the first joint minimax optimization objectives for ensuring harmless LLM outputs under a more realistic and difficult threat model involving a variety of attacks and adaptive adversaries. Our theoretical analysis shows optimizing for our objective is guaranteed to improve robustness, even on unseen instructions and attacks.
- We propose an algorithm, RPO, which can directly optimize for our proposed defense objective with a combination of principled attack selection and discrete optimization.
- The resulting defense, an easily deployable suffix, achieves the state-of-the-art as an effective and universal defense across jailbreaks on JailbreakBench, transfers to closed-source LLMs such as GPT-4, and is resistant to adaptive attacks.

## 2 Related Work

**Adversarial robustness.** A significant body of work in adversarial machine learning studies the inherent susceptibility of neural networks to *adversarial examples* [Szegedy et al., 2014, Goodfellow et al., 2015]. These are inputs designed to be misclassified through perturbations, which include norm-bounded perturbations, small spatial transformations [Xiao et al., 2018], and compositions of transformations [Madaan et al., 2021]. Common defenses to these attacks include input preprocessing [Guo et al., 2018, Nie et al., 2022], distillation [Papernot et al., 2016], provable defenses [Raghunathan et al., 2018, Salman et al., 2020], and adversarial training [Goodfellow et al., 2015, Madry et al., 2018, Tramèr et al., 2018], which has been the most empirically successful. Adversarial training, which is formalized as a minimax optimization [Tu et al., 2019] problem, improves model robustness by optimizing parameters against specially crafted inputs that maximize prediction error.

**Adversarial attacks on LLMs.** Similar attacks have been studied in NLP, including text classification [Ebrahimi et al., 2017, Alzantot et al., 2018, Wang et al., 2022], question-answering [Jia and Liang, 2017], or triggering toxic completions [Wallace et al., 2019, Jones et al., 2023, Zou et al., 2023].

Language models are among the most generally capable models and have been applied to many domains beyond language [Yao et al., 2023, Zhou et al., 2023]. As a result, inducing harmful behaviors has been the primary threat model for LLMs [Carlini et al., 2023]. This has resulted in many recent *jailbreaking attacks*, where an adversary modifies a prompt manually to circumvent alignment training and induce harmful behavior. These attacks can be created manually by humans [Liu et al., 2023b, Wei et al., 2023, Zeng et al., 2024], refined with another LLM [Chao et al., 2023, Mehrotra et al., 2023, Liu et al., 2023a, Jin et al., 2024, Paulus et al., 2024], or generated with discrete optimization [Zou et al., 2023, Lapid et al., 2023, Zhu et al., 2023b, Sadasivan et al., 2024]. In addition, [Huang et al., 2023] finds that simply modifying decoding settings can jailbreak many open-source LLMs. Other attacks include extracting training data [Carlini et al., 2021, Nasr et al., 2023] and misclassification [Zhu et al., 2023a, Wang et al., 2023], but we focus on harmful behaviors.

**Safety and Defenses for LLMs.** Even without an adversary, LLMs are prone to generating biased or toxic content [Sheng et al., 2019, McGuffie and Newhouse, 2020, Deshpande et al., 2023]. To mitigate this, many modern LLMs undergo significant red-teaming [Perez et al., 2022, Mazeika et al., 2024] and additional training such as reinforcement learning with human feedback [Christiano et al., 2017, Ouyang et al., 2022, Bai et al., 2022] to be safer and refuse harmful requests. Additional defenses have recently been proposed with the discovery of additional failure modes, such as jailbreaking, on aligned LLMs. For instance, [Jain et al., 2023] examines simple defenses such as rephrasing the input and finds that the GCG attack [Zou et al., 2023] can be defended with a perplexity filter. Other defenses that have been explored include in-context learning [Zhang et al., 2023], sampling [Li et al., 2023], input processing [Cao et al., 2023, Robey et al., 2023, Kumar et al., 2023], and content moderation [Inan et al., 2023]. While often effective for the threat models considered, many defenses rely on heuristics such as perplexity that can be circumvented by human-readable jailbreaks or require additional inference calls, reducing practicality. Concurrent to our work, [Mo et al., 2024] also considers a similar optimization objective, but only optimizes prompts on the GCG attack, which may limit transferability.

## 3 Towards Adversarial Robustness for LLMs

### 3.1 Attack Objective

In contrast to discriminative models, we are interested in robustness from an *alignment* perspective, in which unwanted behavior can be broader and more harmful than misclassification. We propose a threat model where the adversary can freely select various jailbreaks until the attack is successful, a more challenging and realistic threat model than previous work that only considers one or a few attacks. The only constraints on the adversary are the maximum input length for the LLM, system-level guardrails such as the system prompt, and other special formatting tokens that are inaccessible to users. Otherwise, adversaries can freely modify or add to any accessible part of the input prompt. Consequently, we focus on the *multiattack robustness* setting and aim to create defenses robust to many jailbreaks.

The adversary's goal is to induce an LLM to respond to *any* request, usually harmful ones the model would normally reject. We consider a standard autoregressive language model where a sequence of tokens is mapped to the distribution over the next token. We use $p(\mathbf{y}|\mathbf{x}_{1:n})$ to denote the probability of generating every token in the output sequence $y$ given all previous tokens to that point.

$$p(\mathbf{y} \mid \mathbf{x}_{1:n}) = \prod_{i=1} p(\mathbf{x}_{n+i}|\mathbf{x}_{1:n+i-1}) \tag{1}$$

In the context of jailbreaking, $\mathbf{x}_{1:n}$ is a harmful instruction such as "How do I build a bomb," which we denote as $\hat{\mathbf{x}}_{1:n}$. We consider a modern LLM trained to produce outputs that match human preferences, which is described as a latent reward model $\mathcal{R}^*(\mathbf{y}|\mathbf{x}_{1:n})$ where a high reward is given to outputs more aligned with human evaluations. Thus $\mathcal{R}^*(\mathbf{y}|\hat{\mathbf{x}}_{1:n})$ is high so a vanilla prompt $\hat{\mathbf{x}}_{1:n}$ cannot directly induce the model to respond harmfully.

We consider the setting where the adversary can modify $\hat{\mathbf{x}}_{1:n}$ through various jailbreaks to maximize the probability of producing an output sequence that accepts the harmful request or is toxic. We denote the resulting instruction after a jailbreak as $\tilde{\mathbf{x}}_{1:n}$. In contrast to vision, we do not expect $\tilde{\mathbf{x}}_{1:n}$ to be "stealthy" or semantically equivalent to $\mathbf{x}_{1:n}$, besides the original instruction. The generation process can be formulated as the negative log probability of the target sequences of tokens $\mathbf{y}^\star$ representing the